

NTRO Bitcoin AML Surveillance — Recital Demo Coverage Brief

Every user-visible surface of the working prototype, what it does, and the exact click-path to show it. Read top-to-bottom as the demo run-of-show. **Page 3 guardrails are binding** — several widely-cited figures in the docs do not match the shipped artifacts, and two UI elements are labelled differently than the docs claim.

3

SURFACES

52

FEATURES

15,873

SCORED WALLETS

0.921

GRAPH F1

1 · SYSTEM MAP ARCHITECTURE & SHELL

2 · FEATURE CATALOGUE ALL 52 SURFACES

3 · RUN OF SHOW SCRIPT & GUARDRAILS

↳ API SURFACE

↳ VERIFIED NUMBERS

↳ DO-NOT-CLAIM

What this system is. An air-gapped, offline Bitcoin transaction-traffic surveillance engine for NTRO. It ingests bulk **CSV / JSON / XML** transaction+network exports, correlates the **network layer** (IP, ASN, geolocation via local MaxMind GeoLite2) against the **blockchain layer** (wallet, TXID, input/output graph) in **Neo4j**, then applies ML to cluster entities, flag anomalies, catch **peeling chains** and **CoinJoin mixing**, and score risk — with a full court-admissible explainability trail. Stack: **FastAPI + Celery/Redis + PostgreSQL 16 + Neo4j 5.26 (GDS) + PyTorch 2.4.1 CPU / PyG 2.6.1 + Next.js 16**. Zero cloud egress.

A End-to-End Pipeline — one file in, one court dossier out

NARRATE LEFT — RIGHT; THIS IS THE SPINE OF THE DEMO

01 · UPLOAD Ingest Batch Multipart drop. Format sniffed, streamed to disk, SHA-256 computed as bytes land.	02 · GATE Idempotency Redis file_hash: {sha} check. Duplicate → 409 + original task id. TTL 24 h.	03 · QUEUE Celery Worker process_ingest_file STARTED state, 3,600 s soft limit.	04 · PARSE Normalise CSV, JSON, XML all collapse to one row schema. Downstream never branches on format.	05 · ENRICH GeoIP / ASN MaxMind GeoLite2 City + ASN attach geography and autonomous system to each row.	06 · LOAD COPY → Postgres copy_expert binary streaming. 100k rows in 8.4–12.0 s. No ORM, no row inserts.	07 · SCORE Inline Inference FT-Transformer (18 feats) + peeling/CoinJoin rules + seed proximity + co-spend clustering.	08 · GRAPH Neo4j + Louvain 24,673 :Wallet, 100k :Transaction. Communities → entities.	09 · DETECT Risk Passes Graph Transformer risk (F1 0.9209) + composite calibration into 4 severity bands.	10 · EXPLAIN XAI Artifacts 18-feature attributions, 18×18 attention matrices, GNN subgraphs, evidence trails.	11 · PRESENT Dossier + Sec 65B Alert feed, force graph, forensic drawer, statutory certificate export.
--	--	---	--	--	---	---	--	---	--	---

B Command-Center Shell — orient judges in 20 seconds

- 1

Top Status Bar

Agency identity, **AIR-GAPPED OPERATIONAL** indicator, **wallets-indexed counter** (the "we ingested 100k txs" proof), the **Dual Transformer status chip** (opens the benchmark audit — best single credibility click), and **Ingest Batch**.
- 2

Stream Telemetry **left rail**

Live coverage bar (Scope: **filtered** / **total** + %) that visibly shrinks as you filter, plus a **N Active** badge. Honest, reactive, the easiest "it really works" beat.
- 3

Risk Verdict filter matrix

Four toggle cards with **live counts** — **CRITICAL** **HIGH** **MEDIUM** **LOW**. Clicking Critical collapses the table instantly. Strongest interaction demo.
- 4

Min Anomaly Score

Range slider **0 – 1** with 4 **preset pills** (All **0.25** **0.50** **0.75**) and a live **≥ 0.xx** readout. **No histogram exists** — see guardrail G1.
- 5

Laundering Heuristics

Peeling Chains ("linear multi-hop peel transfers") and **CoinJoin Mixing** ("equal-output mixing rounds") toggles. Demonstrates the deterministic rule layer narrowing the alert set.
- 6

Cluster Partition input

Type a **#{id}** to jump straight to that entity in the topology graph; quick-seed chips included. Bridge from table → graph.
- 7

View tabs

Alerts Stream ↔ Cluster Topology. This switch is the narrative spine of the whole demo: *table → graph correlation*. Switching auto-loads a prime cluster if none is active.
- 8

Alerts search

Address-prefix search, placeholder **Search by Bitcoin address prefix...**, **200 ms debounce**, infinite-scroll offset reset. Shortcut: press **/** or **Ctrl+K** to jump focus mid-sentence.
- 9

The 7 alert columns

Risk Verdict (pill + composite score to 3 dp) · **Entity Address** (truncated, 1-click copy) · **Anomaly Score** (4 dp + micro-bar + percentile chip) · **Cluster Partition** · **Laundering Pattern** · **Intelligence Seed** · **Telemetry Timestamp**.
- 10

Laundering Pattern — 5 signatures, 1 column

Priority chain, first match wins: **N-HOP PEEL** **COINJOIN** **SYBIL CLUSTER** **BEHAVIORAL ANOMALY** **GRAPH CONTAGION**. Great "one column, five laundering signatures" line.
- 11

Infinite scroll + live telemetry

Scroll loads 50 more rows. Footer shows **N of M entities loaded** and a **ticking SYS UTC clock**. Proves the dataset is real, not a 50-row mock.
- 12

Visual language to point at

Monochrome high-stakes light theme, 38 px dense rows, monospace hash truncation. **Risk-tier node colours**: **critical** ≥0.8 · **high** ≥0.6 · **medium** ≥0.4 · **low**.

C API surface

FASTAPI

ROUTE	SERVES
GET /health	Liveness probe
POST /ingest	Async multipart upload → 202 + task id + file hash
GET /ingest/status/{id}	Celery progress, insert / reject counts
POST /ingest/sync/{id}	Post-ingest inline scoring (409 on replay)
GET /api/v1/alerts	Paged, filterable alert feed
GET /api/v1/entity/{addr}/explain	The full forensic dossier payload
GET /api/v1/graph/{cluster}	Bounded subgraph — 150 default, 250 ceiling

D Shipped dataset reality

VERIFIED FROM ARTIFACTS

ARTIFACT	COUNT	SOURCE OF TRUTH
Indexed wallets scored	15,873	composite_risk_scores.json
— by verdict	CR 103 · HI 10 ME 3,112 · LO 12,648	Counted, reproducible
Graph nodes	24,673 :Wallet	wallet_risk_scores.json
Transactions	100,000	synthetic_transactions.csv
Ransomware seeds	11,186	\$1.018B — Ransomwhere, DOI-cited
SHAP attribution sets	4,839 × 18 feats	shap_attributions.json
Attention matrices	4,839	18×18 cross-feature
GNN subgraphs	100	gnn_subgraphs.json — not 500 , see G2

Feature Catalogue — every surface, every click

All remaining prototype features, grouped by surface. Each entry: **what it is** → **how to demo it**. Use exact UI wording in quotes — several labels differ from the internal docs, and misnaming a control on stage reads as unprepared.

- 1 · SYSTEM MAP
- 2 · FEATURE CATALOGUE
- 3 · RUN OF SHOW
- ↳ CLUSTER TOPOLOGY
- ↳ FORENSIC DOSSIER
- ↳ MODALS & CERTIFICATE

E

Cluster Topology — D3 force graph

HIGHEST VISUAL IMPACT

- 13
- Cluster stat badge
- Cluster `#{{id}}` · `N` Nodes · `M` Edges
- Carries a `GNN ACTIVE` flag when explainer highlighting is on: "the graph highlights the exact sub-network the model used."
- 14
- Entity-type filter chips
- ALL High Risk Seeds Wallets TxS IPs — each with a live count. **Click IPs to reveal the network layer** nobody else bothers to show.
- 15
- Graph search + Enter to fly
- Type-to-filter dropdown (top 5); **press Enter and the camera zooms to that node** and pins the inspector. Best "the tool is real" moment.
- 16
- Three layout modes the show-stopper
- Force Physics (default) · Concentric Radar (radial risk rings) · Directed Flow DAG (inflow → outflow columns). **Switching layouts live is the most convincing "we model topology" beat in the demo.**
- 18
- Saliency toggle + viewport controls
- Saliency lights up high-attention edges (≥0.8) — thicker, darker, distinct arrowheads, which visually proves the attention weights are real. Alongside it: **zoom, fit, reset** and **pause-physics** — pause before narrating or nodes drift out from under your sentence.
- 19
- Node-type legend doubles as filter
- Wallet · ■ Tx · ◆ IP Host · ● Seed Entity. Edges: **solid** = value flow, **dashed** = co-spend.
- 20
- Forensic Inspector HUD click any node
- Dark pinned card: severity pill, **risk score /1.0 with bar**, cluster degree, target ID with copy, and on wallet nodes **"Inspect Full Forensic Dossier"**.
- 21
- Multi-head attention breakdown
- Four labelled heads on incident links with α.mean. **Guardrail G4: confirm these are live before narrating them as model output.**

I

Sec 65B / BSA 2023 Statutory Certificate

THE CLOSER

- 41
- Classified document shell
- `SECRET // LAW ENFORCEMENT SENSITIVE` NTR0 letterhead, jurisdiction **Republic of India (BSA 2023 Sec 63 / IEA Sec 65B)**, device UUID, `OFFLINE SOVEREIGN NODE`.
- 42
- Tamper-evident certificate
- Web Crypto **SHA-256** digest on open, copyable **Certificate ID** (NTR0-65B-2026-XXXXXX-XXXX). `SEAL VERIFIED` badge.
- 43
- Statutory findings + Print / Save PDF
- Full Sec 65B / Sec 63 declaration, findings table and model checksums — then a self-contained printable certificate. **Make this your final beat.**

F

NTR0 Forensic Dossier

THE EXPLAINABILITY CORE

- 22
- Risk gauge + verdict + seed badge
- Composite score gauge, verdict pill, and a red intelligence-seed family tag when the wallet is a known threat-intel hit.
- 23
- Automated Forensic Summary
- Pre-written plain-English analyst narrative, generated **deterministically — no LLM in the loop**. Read one sentence aloud: this is your "we refuse to hallucinate evidence" line.
- 24
- Four component score tiles
- Anomaly Risk Rule Bonus Mixing at 5 dp. Kills the black-box objection — the composite is these four parts.
- 25
- Cluster Co-Spending Analysis
- Cluster ID, size, **seed-wallet proximity, anomaly rank percentile** — e.g. "99.82nd percentile". Concrete, quotable, defensible.
- 26
- FT-Transformer Tabular Anomaly
- Reconstruction **MSE to 6 dp**, percentile rank, threshold status against the calibrated boundary `0.0346` — `EXCEEDED` vs `NORMAL`. **Quote 0.0346 aloud; it's a real calibration constant.**
- 27
- Mixing & Laundering Heuristics
- Mixing detected Y/N, peeling-chain status, **mixing hops**, optional **pass-through ratio**, pattern chips.
- 28
- Triggered Rules Audit
- Red chips naming every rule that fired, e.g. `FT_TRANSFORMER_ANOMALY COINJOIN_PATTERN_DETECTED LARGE_CLUSTER`. Rule names map straight back to the code on stage.
- 29
- SHAP Waterfall default view
- Diverging bars from a zero axis — **red** pushes toward an illicit verdict, **green** away. "18 features, every one interrogable."
- 30
- Transformer Attention Matrix 18×18 heatmap
- The transformer's own cross-attention, with live **Mean α** and **peak feature pair**. **Hover 2-3 cells — each returns a plain-English forensic insight string.** Zero effort, huge impact.
- 31
- Topological Subgraph & GNN Explainer
- Neighbour list with type badges and per-node risk scores, plus the explanatory edge set the GNN proved material. "Here is exactly which neighbours drove this score."
- 32
- Downloads
- Download JSON** (raw explain payload) and **Export Certified Legal Dossier (Sec 65B)** → section I. Always end the drill-down here.

G

Ingest Batch modal

ETL PIPELINE

- 33
- Drag-drop dropzone + whitelist
- Accepts `.CSV / .JSON / .XML`, **max 500 MB**, with SHA-256 duplicate validation and air-gapped parsing called out in the dropzone itself.
- 34
- Draggable sample dataset card
- "Sample 2,000 TXs"** — 2,000 rows, 1.1 MB, bundled locally. **Drag the card into the dropzone** rather than hunting for a file. Fully offline-safe by design.
- 35
- Live Celery progress
- Bar climbs to 45% during upload, then the drawer auto-closes on success. Say: "a Celery worker is processing this in the background, and I can poll its state."
- 36
- Resilience states
- 409 duplicate rejection** and an all-rows-duplicate warning with distinct copy. **Demo the 409 on purpose** — it is idempotency proof, not an error. Guardrail G6.
- 37
- Post-ingest inline sync
- Newly uploaded wallets are scored immediately rather than waiting for a full retrain — the "it works on *new data*" answer.

H

Dual Transformer Provenance & Benchmark Audit

OPENS FROM TOP CHIP

- 38
- Four benchmark cards
- Graph F1 0.921** (gate 0.880 ✓, +5.9% over baseline 0.870) · **Peeling Recall 94.8%** (baseline 81.2%) · **Latency 4.8 ms** on CPU (−44%) · **Memory 231 KB** of weights, <120 ms cold boot. **Read all four aloud — it's 10 seconds and it's the most judge-convincing moment in the build.**
- 39
- Head-to-head architecture comparison 5 dimensions
- Active vs baseline vs forensic impact: **tabular engine** (FT-Transformer vs deep MLP AE), **graph engine** (Relational Graph Transformer vs GraphSAGE), **edge relations** (3 heterogeneous vs 1), **explainability** (dual attribution vs SHAP only), **deployment** (231 KB vs 65 KB, zero GPU). Read *one* row fully, not all five.
- 40
- SHA-256 weight provenance
- Full checksums for both `.pt` checkpoints, per-row copy buttons. "These are the exact weight hashes that produced those numbers." **Copy Benchmark Audit Summary** also dumps a formatted block to clipboard.

Run of Show — script, numbers, and guardrails

The right-hand column is **not optional**. The left column is the recommended order. The guardrail column lists claims that are **factually wrong today** — saying them on stage is the fastest way to lose a technical judge.

- 1 · SYSTEM MAP
- 2 · FEATURE CATALOGUE
- 3 · RUN OF SHOW
- ↓ DO-NOT-CLAIM
- ↓ VERIFIED NUMBERS

10
DEMO BEATS

9
GUARDRAILS

0
CLOUD CALLS

J

Recommended 10-beat demo sequence

~8 MINUTES

- 0:00

Cold open — the scale number
Point at the top-bar counter and the telemetry rail. Don't click yet.
*"This is a sovereign, air-gapped NTRO node. It has already indexed **15,873 Bitcoin entities** across **100,000 transactions** — with no internet connection anywhere in the stack."*
- 0:40

Benchmark chip → model credibility
Click **Dual Transformer (FT-Trans + RGT 4-Head) • 4.8ms CPU** in the top bar.
*"Graph F1 **0.921** against a **0.880** promotion gate. Peeling recall **94.8%**. **4.8 milliseconds** on pure CPU, **231 kilobytes** of weights, cold boot under 120 ms. Here are the SHA-256 checksums of the exact weights that produced those numbers."*
- 1:30

Filter the noise down to signal
Click **Risk Verdict** → **Critical**. Watch the coverage bar shrink. Then enable **Peeling Chains** + **CoinJoin Mixing**.
*"From fifteen thousand entities, one click isolates the **103 critical ones** — then the deterministic laundering heuristics narrow it further. This is not a search box; this is a triage funnel."*
- 2:20

Read one row like an analyst
Hover the anomaly bar, then the **Laundering Pattern** badge and the **Intelligence Seed** tag.
*"One row carries: composite risk to three decimals, anomaly reconstruction error with its percentile, the entity's cluster, the **laundering signature** that fired, and whether it's a known ransomware seed."*
- 3:10

Table → graph escalation
Click a row's **Cluster Partition** chip, switch to **Cluster Topology**, then **pause physics**.
*"That score wasn't a black box — it was computed over this exact topology. **150 nodes, 104 edges**, one wallet, one transaction, one IP host, one seed entity."*
- 4:00

Layout theatre + attention proof
Switch **Force** → **Concentric** → **Directed Flow**. Toggle **Saliency** on and off.
"Same graph, three analytical lenses. With saliency on, we're only showing the edges the transformer actually weighted — the ones that moved the risk score."
- 4:50

Node → inspector → full dossier
Click a **seed** node, read the HUD, then **Inspect Full Forensic Dossier**.
*"Composite score **1.000**, critical. Here's the deterministic summary — no LLM, no hallucination. Four component scores, the **0.0346** reconstruction threshold it breached, and the **99.8th percentile rank**."*
- 5:50

The explainability double-punch
Show the **SHAP Waterfall**, then flip to the **Transformer Attention Matrix** and hover 2–3 cells.
"Eighteen features, every one signed and attributed. And this is the transformer's own 18×18 cross-attention — hover any cell and it tells you what that co-dependency means in forensic terms."
- 7:00

Ingest live — and hit the 409 on purpose
Ingest Batch → drag the **Sample 2,000 TXs** card in → watch Celery progress → repeat the drop.
*"Uploaded, fingerprinted with SHA-256, queued to a Celery worker, and scored inline. Now watch — I drop the identical file again and it's **rejected as a duplicate**. That's real idempotency, not a UI guard."*
- 7:40

Close with the legal artefact
Export Certified Legal Dossier (Sec 65B) → read the certificate ID → **Print / Save PDF**.
*"Every finding in this session, sealed under **Section 65B of the Indian Evidence Act and Section 63 BSA 2023**, with a tamper-evident SHA-256 digest bound to this sovereign node."*

K

Do-Not-Claim — binding corrections

FACTS, NOT OPINION

- 1

There is no histogram on the anomaly slider.
The sidebar has a range input and 4 preset pills. Nothing else. Never call it a distribution chart — a judge who opens devtools will catch it.
- 2

Counts are 15,873 — not 17,020. Subgraphs are 100 — not 500.
Internal docs claim 17,020 evidence trails and 500 GNN subgraphs. The shipped JSON artifacts hold **15,873** and **100**. Use the artifact numbers or regenerate first.
- 3

"RISK VECTOR" and "Import Batch" are wrong labels.
The UI says **Risk Verdict** and **Ingest Batch**. Both strings are absent from the source.
- 4

The 4 attention-head bars have hardcoded fallbacks.
Without a live **head_attentions** payload they render constants **0.91 / 0.84 / 0.78 / 0.65**. Same for the 18×18 matrix, which self-generates when absent and shows a **Pre-LN MHSA Saliency** badge. Check the network tab before claiming live weights.
- 5

The top-bar counters have fallback constants.
With no backend, the UI still shows **15,873 / 103 / 10 / 3,112 / 12,648**. Verify the API is live before quoting any count aloud.
- 6

Demo order matters: explore first, ingest last.
A fresh wallet is deliberately **provisional** — explainability panels show honest placeholders. Drill into an already-indexed critical wallet *before* uploading, or the demo looks broken when it's working as designed.
- 7

CoinJoin 100% is in-harness recall, not field accuracy.
The repo explicitly bans inflated claims; published methods sit at **89.2% RF / 87.5% BlockSci**. Peeling **97.2%** recall is the honest figure.
- 8

The Sec 65B dossier is generated client-side.
Web Crypto SHA-256 in the browser. There is **no backend dossier endpoint**. Say "exported from the dashboard", never "generated by our dossier service".
- 9

Don't lead with a convergence story, and don't name the legacy models as production.
FT-Transformer checkpoint records **epochs_trained=1** with zero loss — its evaluation metrics are real, the training log is thin. The autoencoder and GraphSAGE are **legacy fallbacks**; the transformers are what ship.

L

Verified numbers — safe to state

FROM ARTIFACTS + WORK LOGS

METRIC	VALUE	VS BASELINE
Graph risk F1 / AUC	0.9209 / 0.9956	GraphSAGE 0.870
Tabular F1 / AUC	0.6972 / 0.9739	AE baseLine
Combined latency	4.8 ms	8.6 ms → -44%
Peeling recall	97.2% rule · 94.8% TF	81.2% baseline
Ingest throughput	8,307–11,938 tx/s	100k in 8.4–12.0 s
Weight footprint	231 KB	Zero GPU